Lab #: 1
Title: MATLAB and Simulink

Responsible Lab TA: N/A

Group #: Individual Submission
Team members:
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Team Contribution Statement

Team member	Experimental work and data collection	Analysis	MATLAB/Simulink code	Interpretation and validation	Report writing and editing	Overall contribution
Member 1: Temuulen Ankhbayar	TA	TA	TA	TA	TA	100%
Total						100%

Acknowledgement: By submitting this report, all team members confirm that the information above accurately represents their contributions and that all team members have reviewed and approved the final report and associated files.

A common grade will normally be assigned to all team members. The instructor may request additional information when the contribution statement or other evidence indicates substantially unequal participation.

Q1: Simulink Signal Generator and Scope

1. Model description

The model sums a sine wave of amplitude 2 V with a square wave of amplitude 3 V, and forces the output to 1 V whenever the sum lies between 4 V and 5 V inclusive, leaving the output equal to the sum otherwise. Both sources are configured in rad/s, so a setting of 1 rad/s gives a period of 6.28 s and 10 rad/s gives 0.628 s. As Figure 1 shows, two Relational Operator blocks test the sum against Constants of 4 and 5, an AND gate combines them, and that result drives the control port of a Switch block which selects between a Constant of 1 V and the raw sum before the Scope. All three runs use the ode45 variable-step solver with a 1 ms maximum step size and a 20 s stop time.

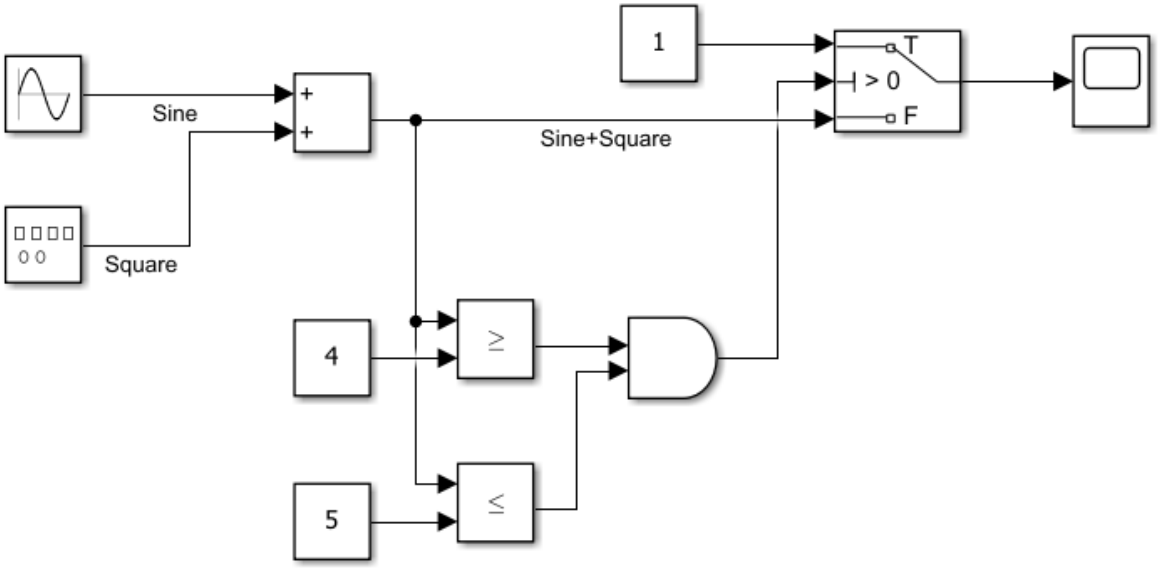


Figure 1: Simulink model. The sum of the sine and square waves is tested against the 4 V and 5 V thresholds, and the result selects between a constant 1 V and the raw sum at the Switch block.

2. Baseline result

With both signals at 1 rad/s the sine and the square wave share a period of 6.28 s, and the square wave begins at its low level of -3 V while the sine is in its positive half cycle, so the two are in antiphase. The square wave therefore holds -3 V while the sine adds at most +2 V, and holds +3 V while the sine subtracts up to 2 V, confining the sum to -3 V to +3 V. The 4 V to 5 V band is never entered, so the conditional branch is never selected, and the output follows the sum everywhere, as Figure 2 shows. This run is a useful reference case precisely because the conditional logic stays inactive, and Figures 3 and 4 show the same model once the band is reached.

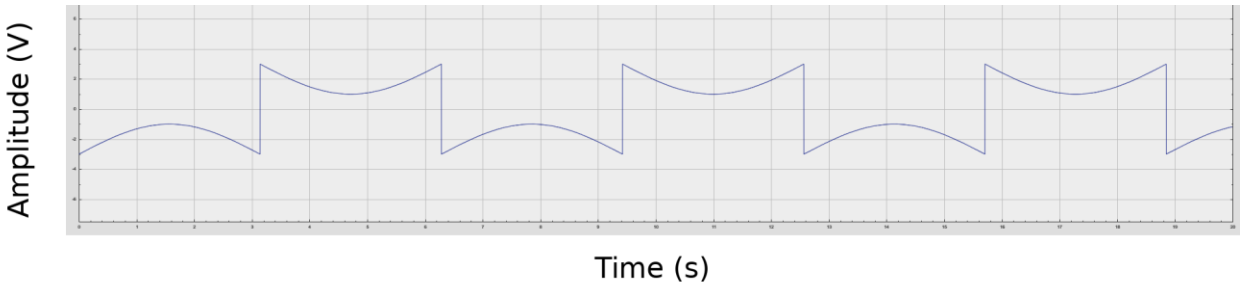


Figure 2: Scope output over 20 s with the sine and the square wave both at 1 rad/s, plotted as Switch output in volts against time in seconds. The sum stays between -3 V and +3 V, so no clipping occurs.

3. Effect of varying frequency

Table 1: Simulation runs used to study the effect of frequency. Stop time was 20 s in every case and the solver settings were held constant so that only the frequencies changed.

Run	Sine frequency (rad/s)	Square frequency (rad/s)	Figure
A (baseline)	1	1	Figure 2
B	10	1	Figure 3
C	1	10	Figure 4

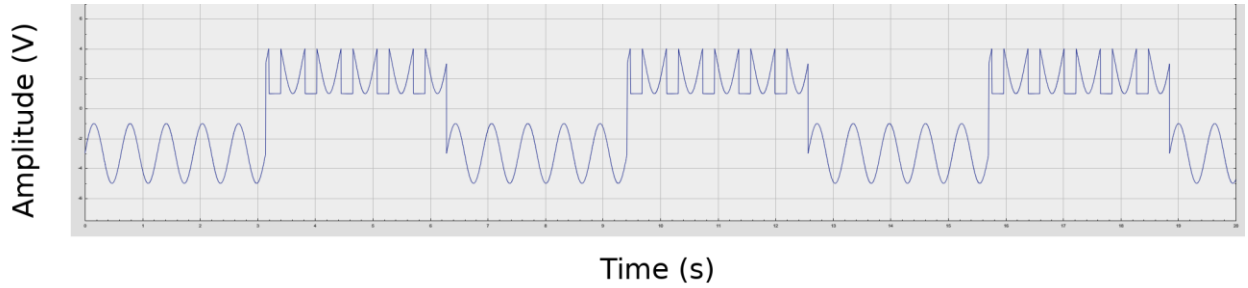


Figure 3: Scope output over 20 s with the sine raised to 10 rad/s and the square wave held at 1 rad/s. Five clipped segments at 1 V appear in each half period during which the square wave holds +3 V, whereas the baseline in Figure 2 produces none.

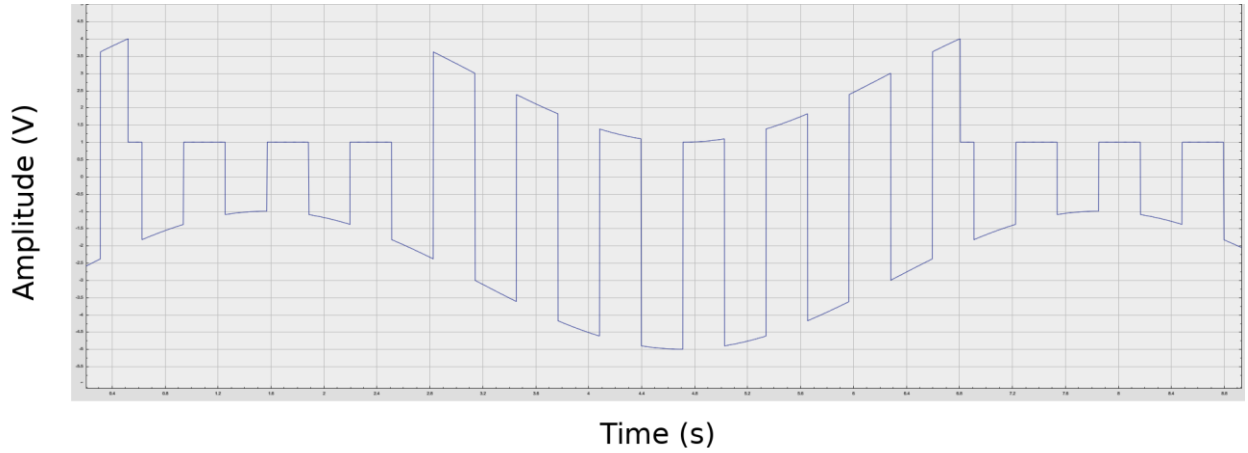


Figure 4: Scope output with the sine held at 1 rad/s and the square wave raised to 10 rad/s, zoomed to the interval 0.4 s to 8.8 s so that the individual pulses remain legible. Clipping occurs only while the slow sine lies above +1 V.

4. Observations

Changing either frequency changes whether the sum ever enters the 4 V to 5 V band, and the three runs in Table 1 show that it is the timing relationship between the two signals, rather than their amplitudes, that decides this. In the baseline of Figure 2 the two waveforms share a period and sit in antiphase, so the square wave holds -3 V exactly while the sine is positive, the sum is confined to -3 V to +3 V, and no clipping occurs. Raising the sine to 10 rad/s in Figure 3 breaks that alignment, because the sine now completes five cycles inside each half period during which the square wave holds +3 V, producing five clipped segments at 1 V in every such half period. Raising the square wave to 10 rad/s instead, in Figure 4, confines clipping to the third of each sine period during which the slow sine lies above +1 V, so the clipped segments appear in short irregular bursts. In both cases the output never exceeds 4 V once the band is active, because the

Switch substitutes 1 V the instant the sum reaches 4 V, which is why the peaks in Figures 3 and 4 are cut flat at 4 V rather than rising to 5 V.

AI tool: Claude (Anthropic Opus5 high).
Purpose: Explanation of Simulink block configuration, and a consistency review of the draft report.
Use in the submitted work: The Simulink model, all simulation runs and all Scope screenshots are my own work. The structure of this report and the wording of the model description and observation sections were drafted with assistance and then edited by me.
Verification: Every numerical claim in the observation section was checked against my own Scope output. The five clipped segments per square wave half period in Figure 3 and the one third clipping window in Figure 4 were confirmed by counting segments on the trace and by comparing the sine and square wave periods computed from their rad/s settings.

References